1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist frames a core question—why RNA-based regulatory programs (m6A-modified lncRNAs and other noncoding RNAs) and tumor metabolic state (one■carbon/mitochondrial metabolism) together determine whether local immune organization leads to effective anti-tumor immunity and therapy benefit in MIBC (e.g., BCG). In other words: why do some tumors with robust local immune structures (TLS/TILs) still fail to respond to immune-based therapy? 2) Tier 1 paths used: I built the hypothesis from the SHMT2 perturbation path (one■carbon/mitochondrial metabolism) and the m6A-related lncRNA (AL161431.1) path because they provide a clear, testable mechanism linking tumor metabolic flux to epitranscriptomic stabilization of oncogenic lncRNA programs that could suppress effective anti-tumor immunity. I considered DDIT4 important (links immune stimulation to stress/mTOR), but kept it out of the core pair to preserve a focused two■feature mechanistic claim. 3) Tier 2 evidence that supports the claim: PTEN/mTOR axis (PTEN) and stress-response signaling (DDIT4) intersect with metabolic control and immune signaling, providing plausible modulatory nodes; MAFG■AS1, circMMP9 and HRG1 indicate complementary RNA regulatory and redox/metabolic routes that could modify immune/therapy outcomes. These Tier 2 nodes justify biological plausibility and provide candidate covariates for follow-up stratified analyses. 4) Why I chose therapy_response as the primary endpoint: The Ontologist explicitly linked DDIT4 to BCG and HRG1 to cisplatin effects, and the central puzzle concerns why immune activation (BCG/TLS) does or does not translate to clinical benefit—thus a therapy response proxy (BCG responsiveness) is the most directly relevant, actionable clinical endpoint for translational follow-up. 5) What I deliberately excluded and why: I excluded DDIT4 from the 1–2 anchor genes to keep the core hypothesis tightly focused on metabolic (SHMT2) and epitranscriptomic (AL161431.1) drivers of immune evasion; DDIT4 and other Tier 2 genes were moved to secondary_modifiers to be tested as effect modifiers in downstream analyses. I also excluded Tier 3 exploratory nodes (ALK, β3GnT5, large gene lists) from the main hypothesis because they are broader, less directly connected, and would dilute mechanistic specificity?

Working Hypothesis

In muscle invasive bladder cancer (MIBC) treated with intravesical BCG, tumors that exhibit TLS-like lymphoid aggregates and harbor high activity of mitochondrial one-carbon metabolism (high SHMT2) together with high expression of the m6A-stabilized oncogenic lncRNA AL161431.1 form a distinct immunosuppressive/metabolically reprogrammed phenotype. This AL161431.1↑/SHMT2↑ state, supported by TLS-like organization, blunts antigen presentation and effector T cell function, leading to reduced and less durable BCG responses. Clinically, patients with TLS-high AND SHMT2-high AND AL161431.1-high tumors will have poorer therapy_response to BCG compared with TLS-high tumors lacking this metabolic–epitranscriptomic signature. DDIT4 and other Tier2 modifiers (PTEN, MAFG-AS1, HRG1, circMMP9) may modulate the magnitude of this effect and serve as covariates or co-targets in combination strategies.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	AL161431.1, SHMT2
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

AL161431.1: AL161431.1) path because they provide a clear, testable mechanism linking tumor metabolic flux to epitranscriptomic stabilization of oncogenic lncRNA programs that could suppress effective anti-tumor immunity.

SHMT2: SHMT2 perturbation path (one■carbon/mitochondrial metabolism) and the m6A-related lncRNA (AL161431.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	DDIT4	Supporting modifier from knowledge graph
Secondary	PTEN	Supporting modifier from knowledge graph
Secondary	MAFG-AS1	Supporting modifier from knowledge graph
Secondary	HRG1	Supporting modifier from knowledge graph
Secondary	circMMP9	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: prospective or retrospective cohorts with TLS quantification via multiplex IHC and spatial transcriptomics, assessment of SHMT2 and AL161431.1 expression (and surrogate m6A status via METTL3/14 signaling), and correlation with BCG response;

Prediction 2: in vitro co-culture assays in bladder cancer models manipulating SHMT2 and AL161431.1 to evaluate antigen presentation markers (MHC I/II, TAP1), adenosine/nucleotide metabolism, and T cell activation;

Prediction 3: in vivo models combining BCG with SHMT2 or AL161431.1 perturbations to assess TLS formation, immune infiltrates, and response durability;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- DDIT4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PTEN: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.